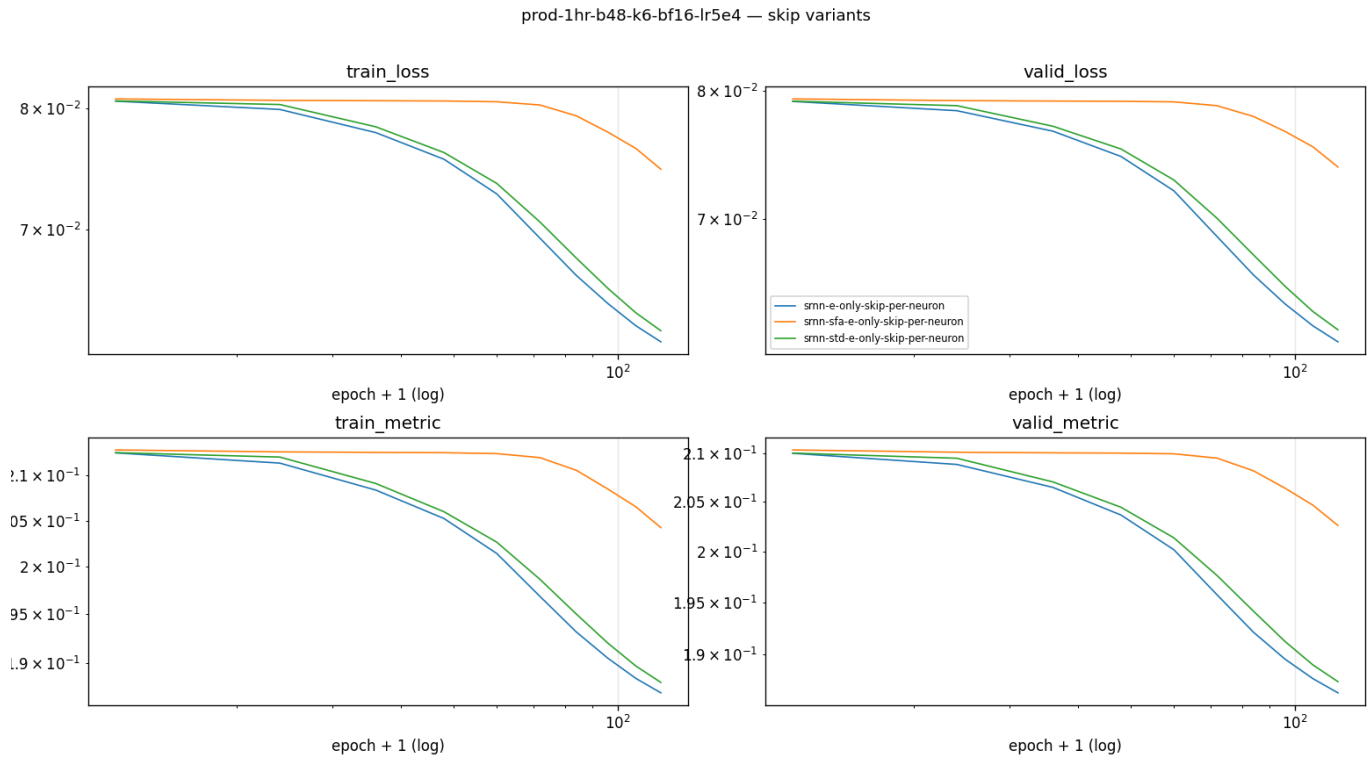


Run: prod-1hr-b48-k6-bf16-lr5e4

Log-log loss curves



Final test_loss (init - last) [skip]:
srnn-e-only-skip-per-neuron: 0.07877 → 0.05990
srnn-sfa-e-only-skip-per-neuron: 0.09420 → 0.07245
srnn-std-e-only-skip-per-neuron: 0.07907 → 0.06061

Figure 1: skip variants — log-log loss/metric

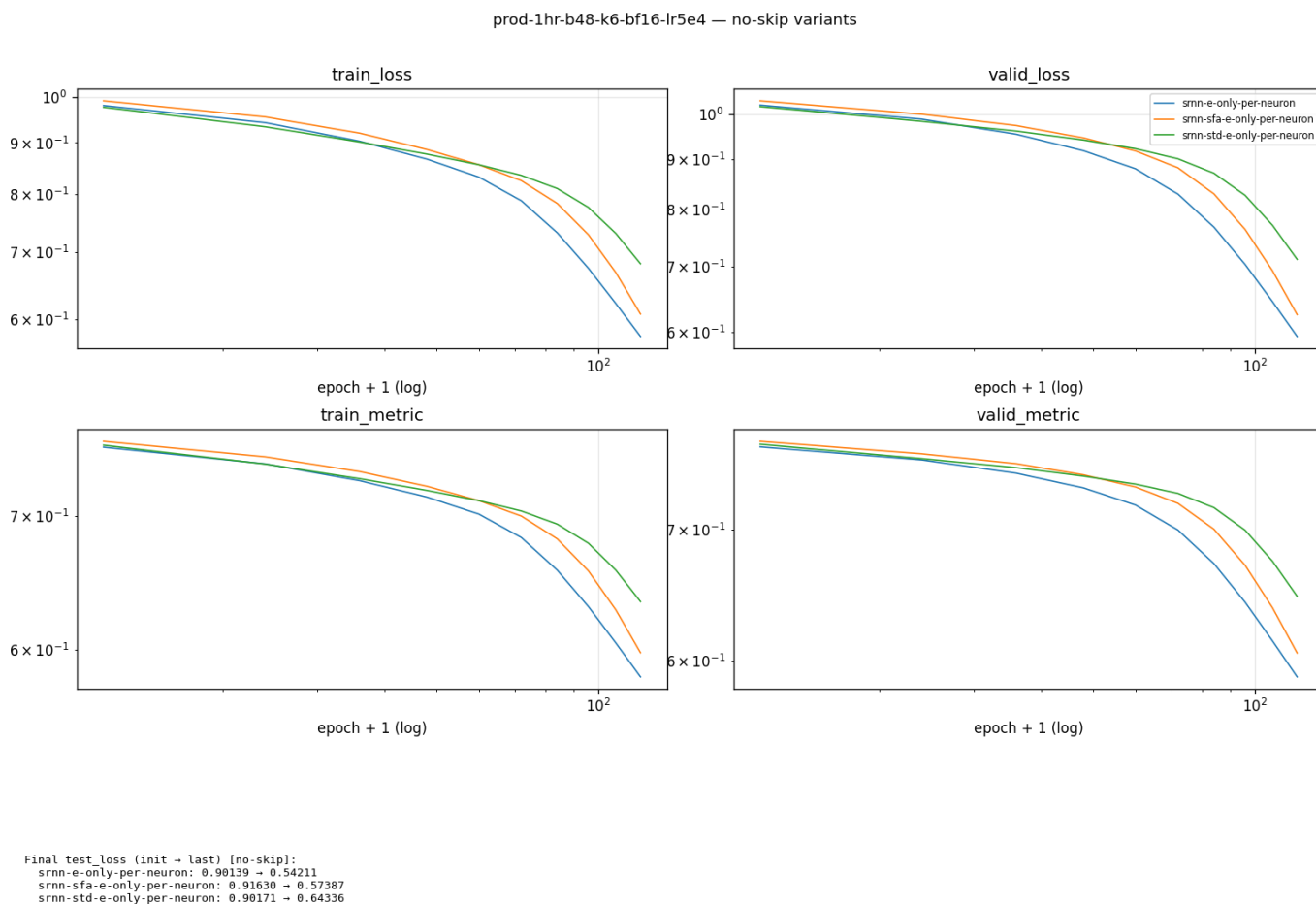


Figure 2: no-skip variants — log-log loss/metric

srnn-e-only-per-neuron

prod-1hr-b48-k6-bf16-lr5e4 — effective time constants (srnn-e-only-per-neuron)

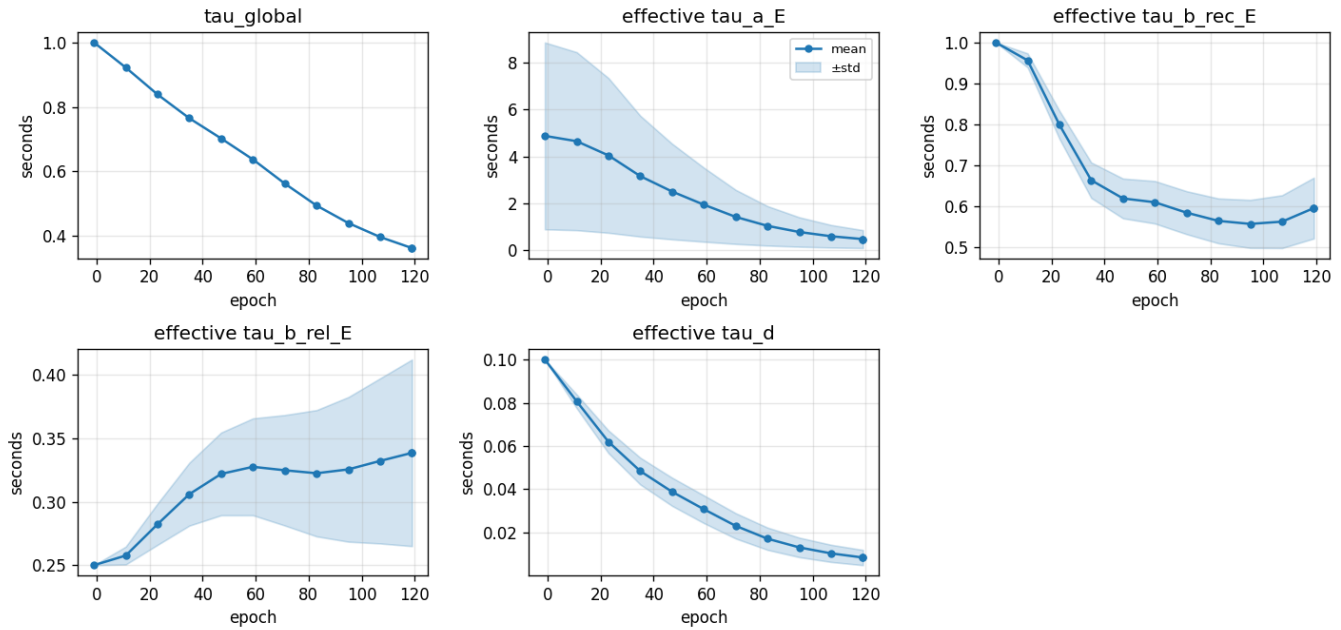
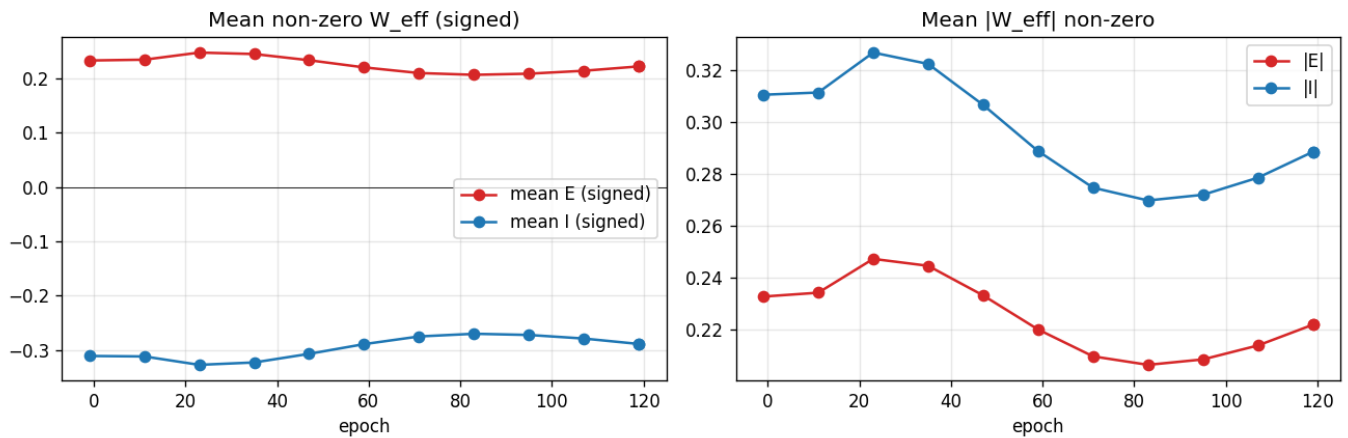


Figure 3: Tau evolution

prod-1hr-b48-k6-bf16-lr5e4 — recurrent E vs I weights (srnn-e-only-per-neuron)

Figure 4: W_{EI} evolution

srnn-e-only-per-neuron — parameter table

```
# prod-1hr-b48-k6-bf16-lr5e4 - variant srnn-e-only-per-neuron (k=0)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.36238	-	-	scalar
tau_d (s)	+0.1	+0.008386	0	0.003537	300
tau_a_E (s)	+4.8777	+0.46351	3.996	0.3839	450
tau_b_rec_E (s)	+1	+0.59595	0	0.07419	150
tau_b_rel_E (s)	+0.25	+0.33863	0	0.07362	150
W_eff (E src, signed)	+0.23267	+0.22179	0.07641	0.08375	15048
W_eff (I src, signed)	-0.31062	-0.28856	0.07724	0.08292	14977
W_eff (E src)	+0.23267	+0.22179	0.07641	0.08375	15048
W_eff (I src)	+0.31062	+0.28856	0.07724	0.08292	14977
W_in_eff (input neurons)	-0.0003386	+0.00077924	0.1012	0.04677	6675
c_E (SFA coupling)	+0.05	+0.14027	0	0.05316	450
c_0_E (SFA offset)	+0	-0.8618	0	0.218	450
a_0 (threshold)	+0.35	-0.4167	0	0.1519	300
readout_weight	-2.8854e-05	-0.0042568	0.06604	0.2784	6675
readout_bias	+0	+0.0055984	0	0.2357	89
ic.ic (per-variant init)	+0.10386	+0.10386	0.3828	0.3828	900

srnn-e-only-skip-per-neuron

prod-1hr-b48-k6-bf16-lr5e4 — effective time constants (srnn-e-only-skip-per-neuron)

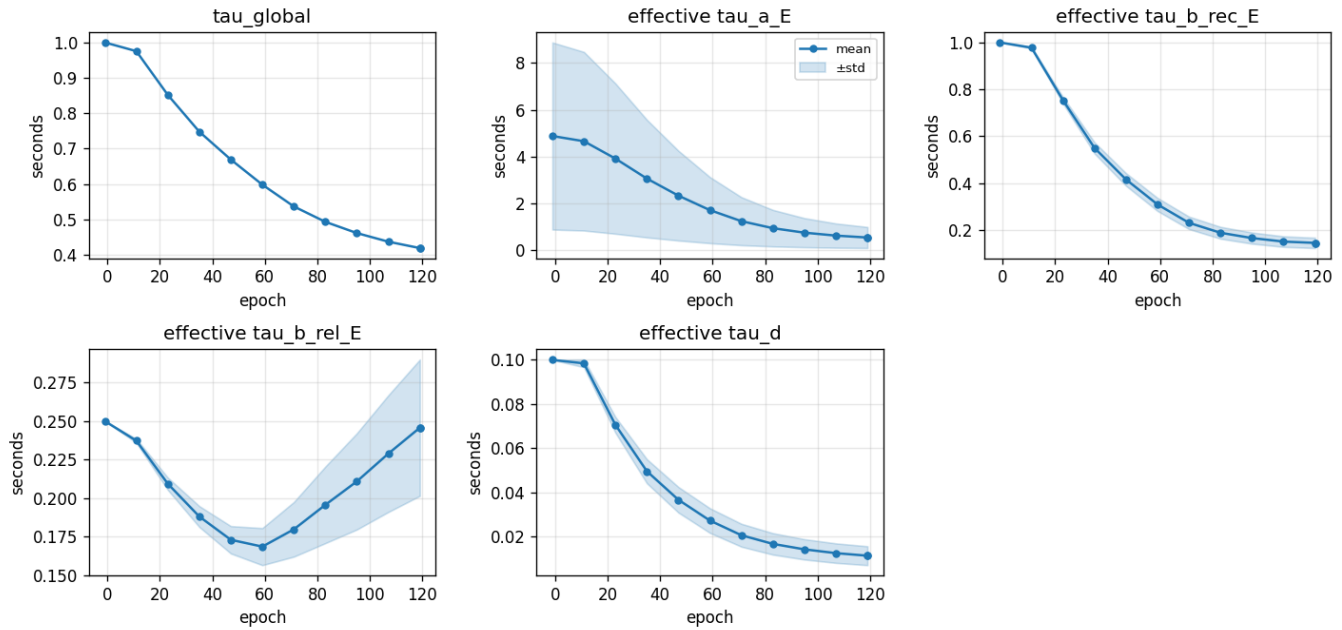
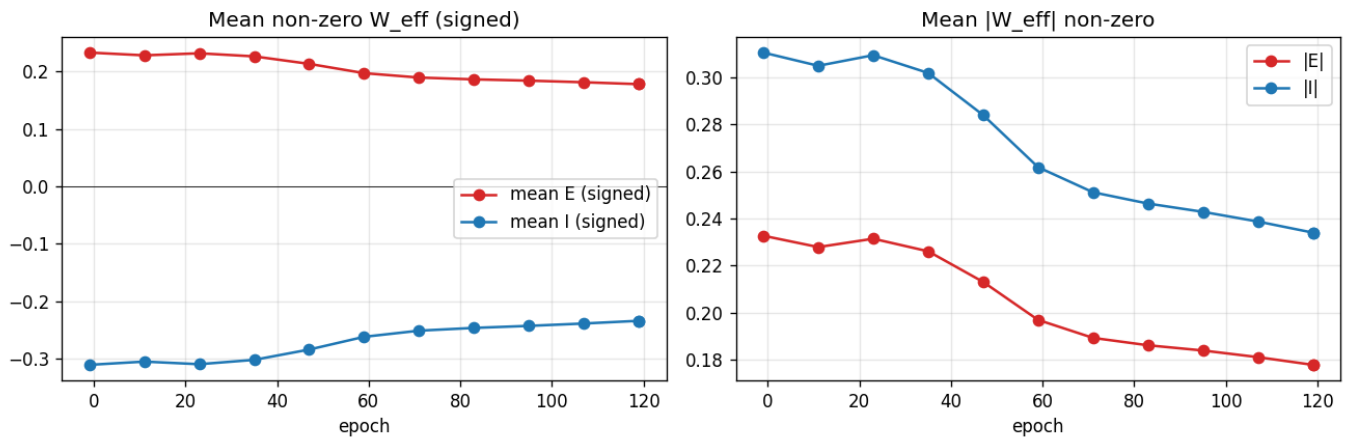


Figure 5: Tau evolution

prod-1hr-b48-k6-bf16-lr5e4 — recurrent E vs I weights (srnn-e-only-skip-per-neuron)

Figure 6: W_{EI} evolution

srnn-e-only-skip-per-neuron — parameter table

prod-1hr-b48-k6-bf16-lr5e4 - variant srnn-e-only-skip-per-neuron (k=1)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.41895	-	-	scalar
tau_d (s)	+0.1	+0.011451	0	0.004285	300
tau_a_E (s)	+4.8777	+0.54239	3.996	0.4506	450
tau_b_rec_E (s)	+1	+0.14509	0	0.02215	150
tau_b_rel_E (s)	+0.25	+0.24571	0	0.04421	150
W_eff (E src, signed)	+0.23267	+0.17774	0.07641	0.06325	15048
W_eff (I src, signed)	-0.31062	-0.234	0.07724	0.06502	14977
W_eff (E src)	+0.23267	+0.17774	0.07641	0.06325	15048
W_eff (I src)	+0.31062	+0.234	0.07724	0.06502	14977
W_in_eff (input neurons)	-0.00084237	-0.00033092	0.1008	0.03457	6675
c_E (SFA coupling)	+0.05	+0.15993	0	0.06398	450
c_0_E (SFA offset)	+0	-0.37073	0	0.1716	450
a_0 (threshold)	+0.35	-0.078787	0	0.1262	300
readout_weight	+0.00078219	-0.0028954	0.06654	0.1552	6675
readout_bias	+0	+0.12476	0	0.1602	89
ic.ic (per-variant init)	+0.10327	+0.10327	0.3829	0.3829	900

srnn-sfa-e-only-per-neuron

prod-1hr-b48-k6-bf16-lr5e4 — effective time constants (srnn-sfa-e-only-per-neuron)

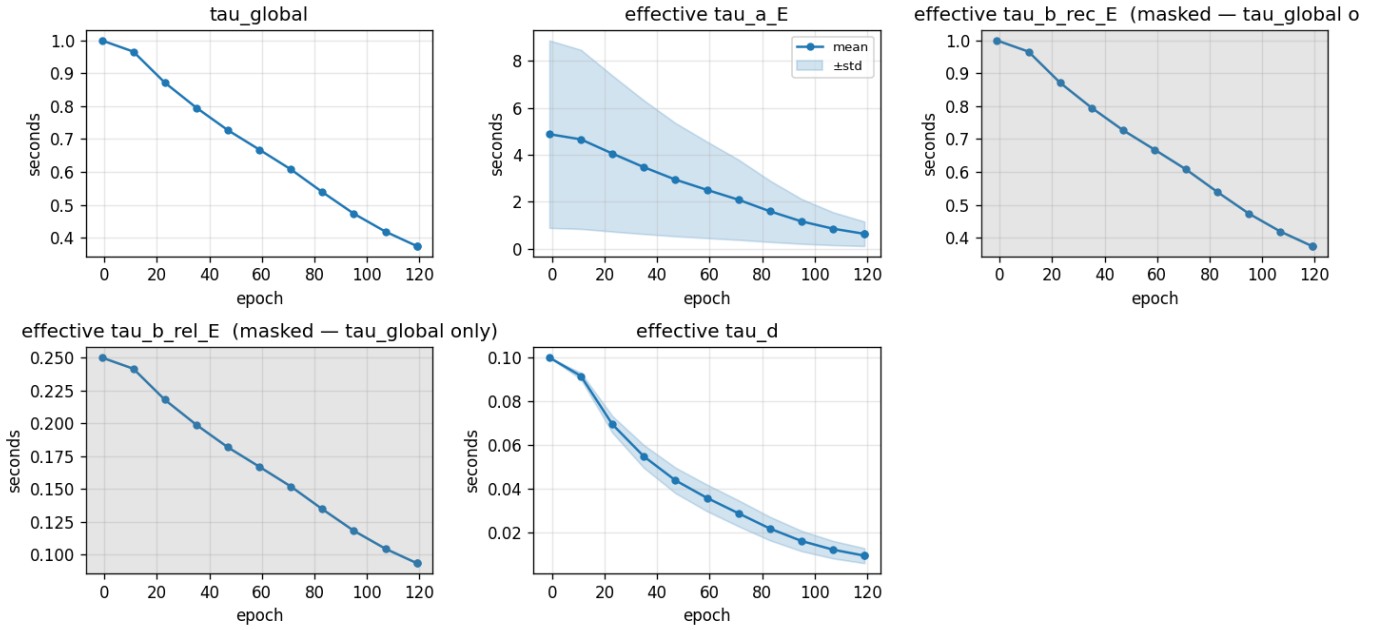
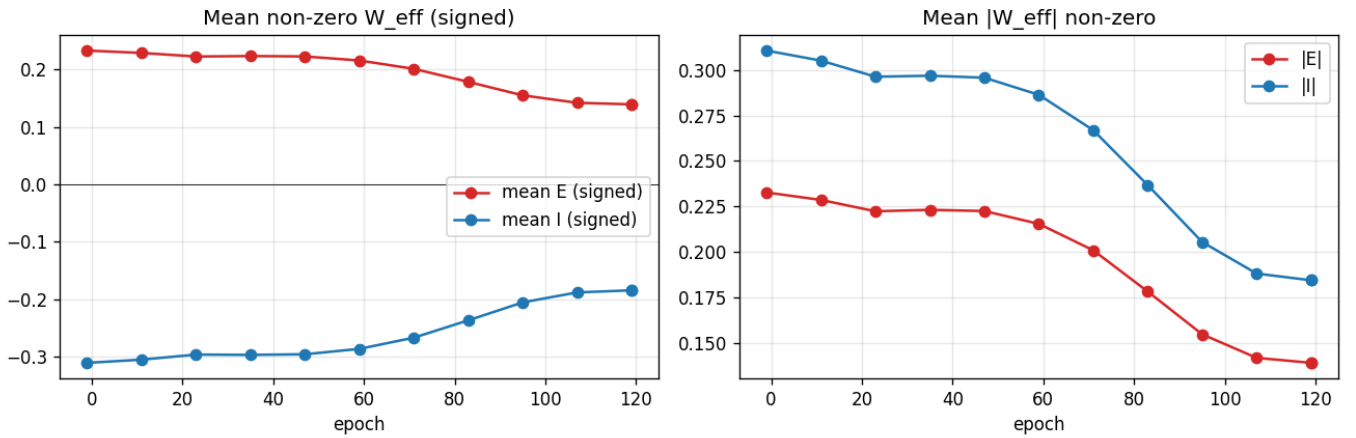


Figure 7: Tau evolution

prod-1hr-b48-k6-bf16-lr5e4 — recurrent E vs I weights (srnn-sfa-e-only-per-neuron)

Figure 8: W_{EI} evolution

srnn-sfa-e-only-per-neuron — parameter table

```
# prod-1hr-b48-k6-bf16-lr5e4 - variant srnn-sfa-e-only-per-neuron (k=2)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.37433	-	-	scalar
tau_d (s)	+0.1	+0.0092731	0	0.003436	300
tau_a_E (s)	+4.8777	+0.63029	3.996	0.5223	450
tau_b_rec_E (s)	+1	+0.37433	0	0	150
tau_b_rel_E (s)	+0.25	+0.093583	0	0	150
W_eff (E src, signed)	+0.23267	+0.13908	0.07641	0.04858	15048
W_eff (I src, signed)	-0.31062	-0.18443	0.07724	0.05026	14977
W_eff (E src)	+0.23267	+0.13908	0.07641	0.04858	15048
W_eff (I src)	+0.31062	+0.18443	0.07724	0.05026	14977
W_in_eff (input neurons)	-0.00015382	-0.0016668	0.09934	0.0522	6675
c_E (SFA coupling)	+0.05	+0.094336	0	0.02552	450
c_0_E (SFA offset)	+0	-0.34523	0	0.1735	450
a_0 (threshold)	+0.35	+0.19378	0	0.09833	300
readout_weight	-0.00098972	+0.0099345	0.06678	0.1995	6675
readout_bias	+0	-0.074786	0	0.1139	89
ic.ic (per-variant init)	+0.13078	+0.13078	0.4172	0.4172	900

srnn-sfa-e-only-skip-per-neuron

prod-1hr-b48-k6-bf16-lr5e4 — effective time constants (srnn-sfa-e-only-skip-per-neuron)

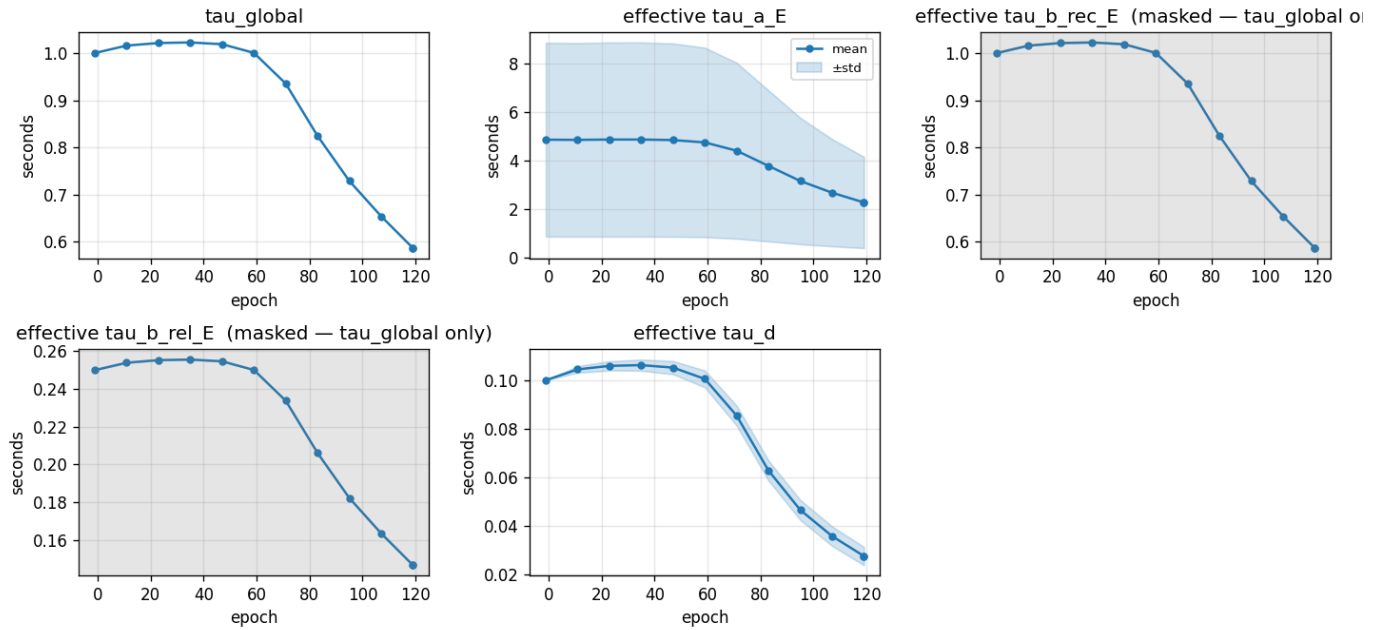
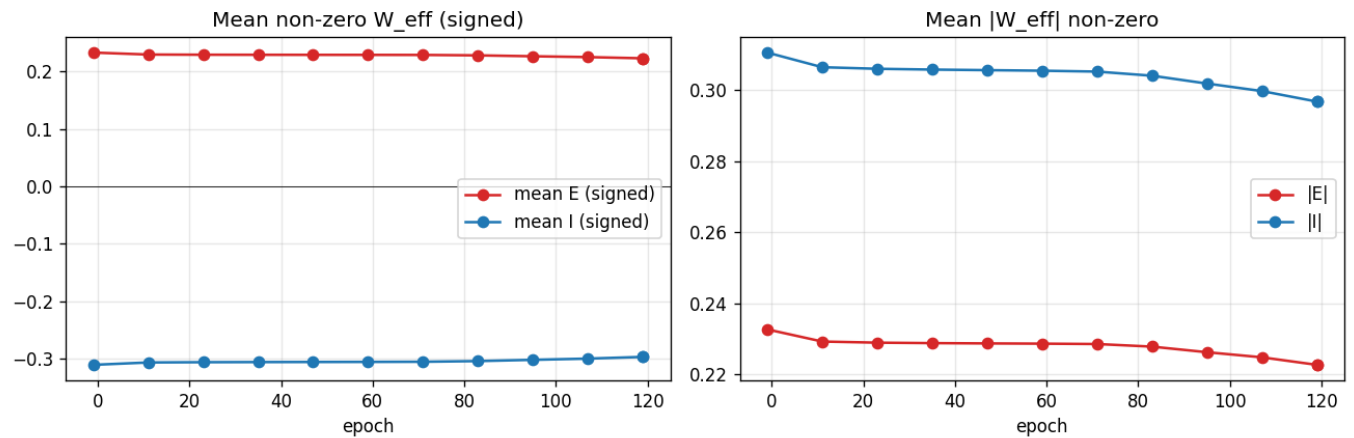


Figure 9: Tau evolution

prod-1hr-b48-k6-bf16-lr5e4 — recurrent E vs I weights (srnn-sfa-e-only-skip-per-neuron)

Figure 10: W_{EI} evolution

srnn-sfa-e-only-skip-per-neuron — parameter table

prod-1hr-b48-k6-bf16-lr5e4 - variant srnn-sfa-e-only-skip-per-neuron (k=3)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.58665	-	-	scalar
tau_d (s)	+0.1	+0.027612	0	0.003754	300
tau_a_E (s)	+4.8777	+2.2891	3.996	1.882	450
tau_b_rec_E (s)	+1	+0.58665	0	0	150
tau_b_rel_E (s)	+0.25	+0.14666	0	0	150
W_eff (E src, signed)	+0.23267	+0.22269	0.07641	0.07435	15048
W_eff (I src, signed)	-0.31062	-0.29682	0.07724	0.07562	14977
W_eff (E src)	+0.23267	+0.22269	0.07641	0.07435	15048
W_eff (I src)	+0.31062	+0.29682	0.07724	0.07562	14977
W_in_eff (input neurons)	-0.0011621	-0.0024476	0.09962	0.0656	6675
c_E (SFA coupling)	+0.05	+0.054701	0	0.008876	450
c_0_E (SFA offset)	+0	-0.00043644	0	0.1036	450
a_0 (threshold)	+0.35	+0.34929	0	0.08355	300
readout_weight	+0.00016801	-0.0032955	0.0673	0.07105	6675
readout_bias	+0	+0.045539	0	0.05694	89
ic.ic (per-variant init)	+0.09231	+0.09231	0.5294	0.5294	900

srnn-std-e-only-per-neuron

prod-1hr-b48-k6-bf16-lr5e4 — effective time constants (srnn-std-e-only-per-neuron)

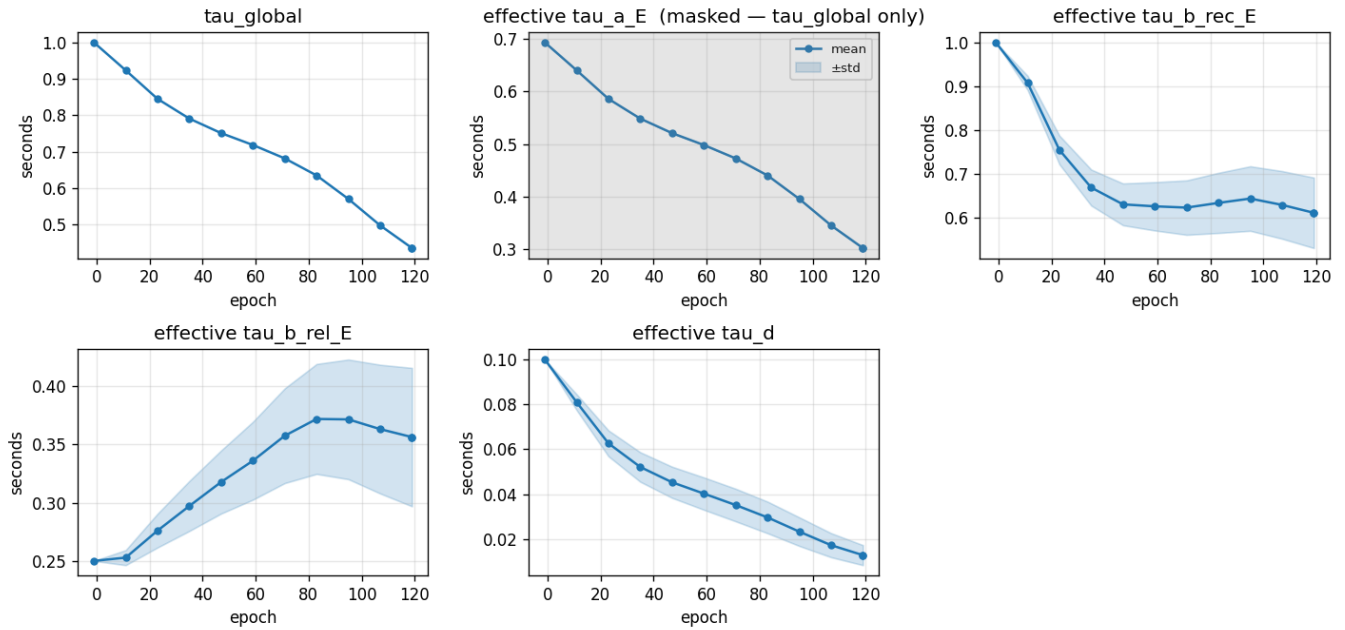
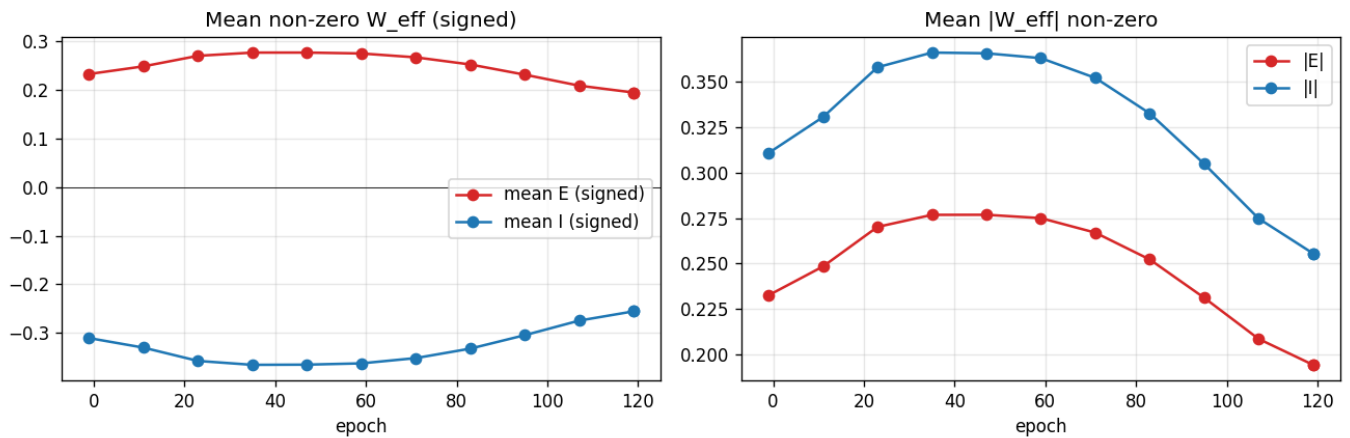


Figure 11: Tau evolution

prod-1hr-b48-k6-bf16-lr5e4 — recurrent E vs I weights (srnn-std-e-only-per-neuron)

Figure 12: W_{EI} evolution

srnn-std-e-only-per-neuron — parameter table

prod-1hr-b48-k6-bf16-lr5e4 - variant srnn-std-e-only-per-neuron (k=4)
Effective values (post-transform). Means/stds computed over the indicated dimension.

effective parameter	init mean	final mean	init std	final std	n
tau_global (s)	+1	+0.43615	-	-	scalar
tau_d (s)	+0.1	+0.012852	0	0.004471	300
tau_a_E (s)	+0.69315	+0.30232	0	0	450
tau_b_rec_E (s)	+1	+0.6102	0	0.08069	150
tau_b_rel_E (s)	+0.25	+0.35639	0	0.05948	150
W_eff (E src, signed)	+0.23267	+0.19455	0.07641	0.07003	15048
W_eff (I src, signed)	-0.31062	-0.25558	0.07724	0.07187	14977
W_eff (E src)	+0.23267	+0.19455	0.07641	0.07003	15048
W_eff (I src)	+0.31062	+0.25558	0.07724	0.07187	14977
W_in_eff (input neurons)	+0.00019944	-0.00098555	0.09994	0.06439	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	-0.28501	0	0.129	300
readout_weight	+0.00074661	+0.0058971	0.06657	0.23	6675
readout_bias	+0	-0.090532	0	0.2547	89
ic.ic (per-variant init)	+0.090819	+0.090819	0.3835	0.3835	900

srnn-std-e-only-skip-per-neuron

prod-1hr-b48-k6-bf16-lr5e4 — effective time constants (srnn-std-e-only-skip-per-neuron)

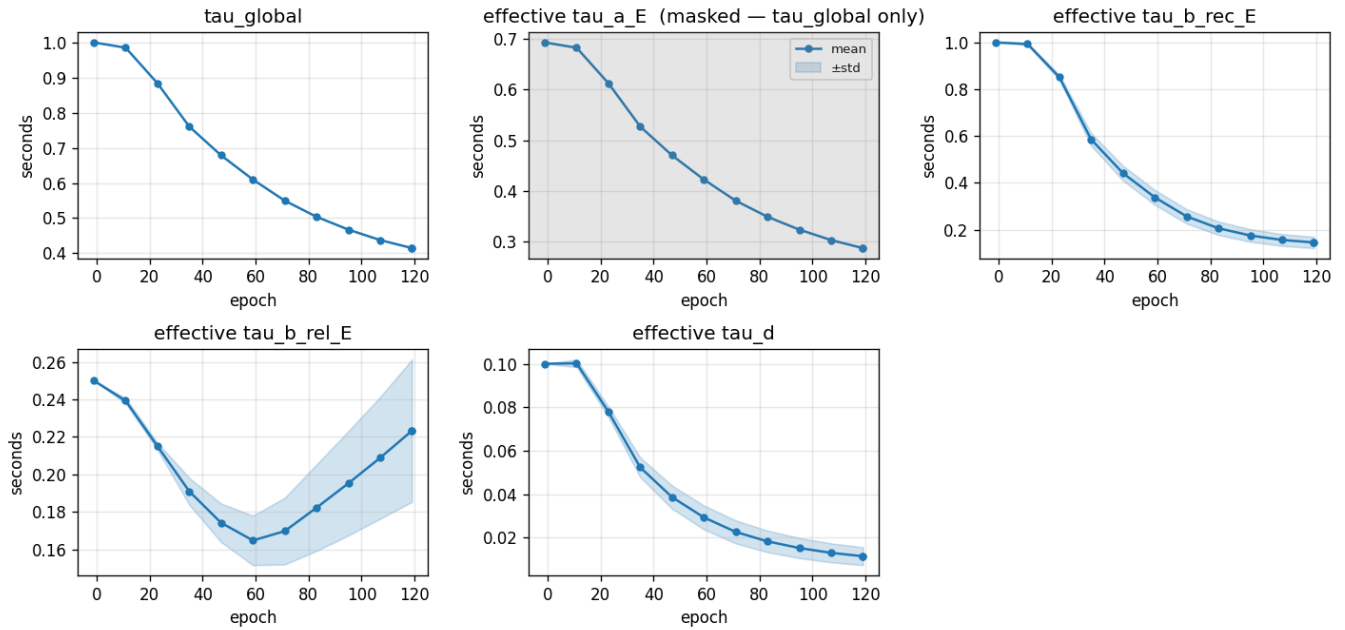


Figure 13: Tau evolution

prod-1hr-b48-k6-bf16-lr5e4 — recurrent E vs I weights (srnn-std-e-only-skip-per-neuron)

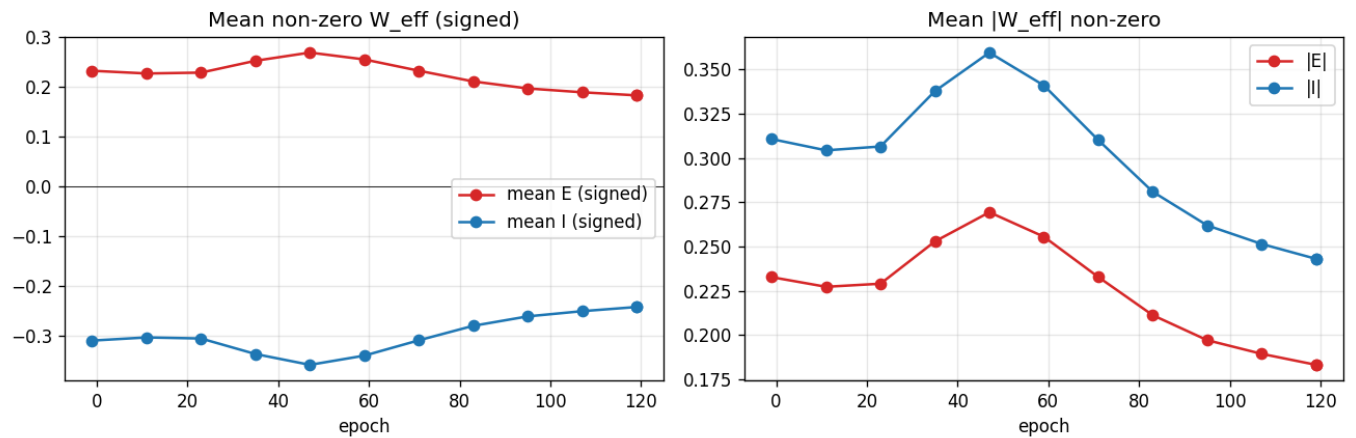


Figure 14: W_EI evolution

srnn-std-e-only-skip-per-neuron — parameter table

```
# prod-1hr-b48-k6-bf16-lr5e4 - variant srnn-std-e-only-skip-per-neuron (k=5)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.41461	-	-	scalar
tau_d (s)	+0.1	+0.011418	0	0.004133	300
tau_a_E (s)	+0.69315	+0.28739	0	0	450
tau_b_rec_E (s)	+1	+0.14509	0	0.0241	150
tau_b_rel_E (s)	+0.25	+0.2233	0	0.03794	150
W_eff (E src, signed)	+0.23267	+0.18315	0.07641	0.06494	15048
W_eff (I src, signed)	-0.31062	-0.24298	0.07724	0.06681	14977
W_eff (E src)	+0.23267	+0.18315	0.07641	0.06494	15048
W_eff (I src)	+0.31062	+0.24298	0.07724	0.06681	14977
W_in_eff (input neurons)	+0.00036119	+0.00034617	0.09956	0.03285	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	+0.0086861	0	0.1454	300
readout_weight	+0.00055372	-0.0046403	0.06572	0.1426	6675
readout_bias	+0	-0.0056223	0	0.1422	89
ic.ic (per-variant init)	+0.083543	+0.083543	0.384	0.384	900